Implementation Guidelines

20MAD481/482 MINI PROJECT (COMPUTATIONAL MATHEMATICS)

This document outlines the standard operating procedures, evaluation rubrics, and strategic action plan for executing the 7th-Semester CSE Mini Project. These guidelines strictly adhere to the 2024 syllabus while providing actionable strategies to manage the computational nature of the coursework alongside the students' existing major project workloads.

1. Administrative Structure

Evaluation is divided into Continuous Internal Evaluation (CIE) and End Semester Examination (ESE). Two distinct committees must be established to oversee this process:

- **Project Evaluation Committee (PEC) [For CIE]:** Composed of the Head of Department (or senior faculty/Chairman), the Mini Project Coordinator, and the respective Project Supervisor.
- **End Semester Examination Committee [For ESE]:** Composed of the HoD/Chairman, an **External Expert** (Industry/Research/Academia), and the Project Coordinator. (Supervisor presence is desirable).

2. Implementation Timeline & Action Plan

To ensure steady progress over the 14-week semester without overwhelming students, supervisors and teams must adhere to the following structured timeline:

Weeks 1 - 2 ***Initiation & Ideation***	**Team Formation:** Max 4 students per group. **Domain Selection:** Focus on domains translating computational mathematics to code (e.g., Image Processing, Graph Theory in AI, Optimization). Python, NumPy, and Matplotlib are highly recommended to minimize the learning curve.
Week 3 ***Proposal Submission***	Submit detailed Project Proposal including objective, scope, resources, and a formal Work Breakdown Structure (WBS). *Supervisor awards up to 5 marks for scheduling here.*
Weeks 4 - 6 ***Execution Begins***	Data collection, mathematical modeling, and initial coding. **Mandatory:** Establish a strict 5-minute weekly window during project hours for the Supervisor to sign the students' Daily/Weekly Activity Diaries.

Week 7 ***Interim Evaluation***	**PEC Review (15 Marks):** Focus on objectives, methodology, literature review, and progress. Presentations should be concise (5-7 slides) focusing on the math and preliminary algorithms.
Weeks 8 - 12 ***Implementation & Draft***	Finalizing the code and evaluating results against mathematical metrics. **Supervisor Validation:** Conduct a live code run to verify results and objectives are met *(Supervisor awards 5 marks for Completion)*. Draft report approved by supervisor must be submitted.
Week 13 ***Final Evaluation***	**PEC Review (15 Marks):** Focus on attainment of deliverables, quality of results, and interaction. Students have **5 working days** to incorporate PEC modifications into the report.
Week 14 - 15 ***ESE Submission***	Submission of final Project Reports (duly certified by HoD). ESE Committee evaluates technical content and presentation (75 Marks).

3. Complete Evaluation Scheme (150 Marks)

A minimum of 75 marks is required to pass. The evaluation is split evenly between Internal and External reviews.

Evaluation Phase	Evaluator	Component / Focus Area	Marks
CIE (75 Marks)	Project Supervisor	Daily Log (10), Lit Survey (5), Scheduling (5), Completion/Live Demo (5)	25
	PEC	**Interim Review:** Tech Content (7.5), Presentation (7.5)	15
	PEC	**Final Review:** Tech Content (7), Presentation (8)	15
	PEC	**Mini Project Report:** Quality, Content, Formats, References	20
ESE (75 Marks)	ESE Committee	**Technical Content**	60
	ESE Committee	**Presentation**	15

Strategic Notice: The Daily/Weekly Logbook

The daily/weekly activity diary accounts for **10 marks** under the Supervisor's CIE. To prevent reverse-engineering of logs at the semester's end, Supervisors must sign these diaries weekly. The diary must reflect day-to-day observations, data gathered, and student reasoning.

4. Final Report & Formatting Directives

To secure the 8 marks dedicated specifically to "Templates/Formats followed" and "Adequacy of relevant references" in the PEC Report evaluation, the following directives apply:

- **LaTeX Template Mandate:** Given the heavy mathematical nature of the project, all final reports should ideally be typeset in LaTeX using the department's provided template. This ensures professional formatting of equations and automates reference adequacy (via BibTeX).
- **Mandatory Structure:** As per syllabus guidelines, the report *must* include the following sections exactly:
  1. Executive Summary
  2. Introduction
  3. Methodology
  4. Results
  5. Discussion
  6. Conclusion
  7. Recommendations

5. Role-Specific Actionable Guidelines

For Students:

- **Leverage Existing Skills:** Do not spend weeks learning a new programming language. Use Python and standard data science libraries (NumPy, PyTorch, SciPy) to implement your mathematical models quickly.
- **Logbook Discipline:** Bring your logbook to every 4-hour project session. Record your coding hurdles, mathematical insights, and supervisor feedback. A well-maintained log is an easy 10 marks.
- **Live Demonstrations:** Be prepared to run your algorithms live during the Week 12/13 check-in. Static screenshots will not suffice for the "Completion of Project" marks.

For Supervisors:

- **Enforce the WBS:** Do not allow students to begin coding without a mathematically sound and computationally feasible Work Breakdown Structure.
- **Batch Logbook Signings:** Dedicate the first or last 10 minutes of the weekly project block exclusively to reviewing and signing student diaries to maintain continuous evaluation momentum.
- **Technical Verification:** During final reviews, focus on the *interpretation* of the data (CO3). Did their custom algorithm perform better? Ensure students understand the math behind their code, not just the outputs.